Supplementary Material for "Neural Belief Corrections for MPC: Plan-Consistent Scalar Budgets versus Myopic Directional Constraints"

Thanana Nuchkrua, Xiaoqi Liu, and Sudchai Boonto

This document supports the eight-page Technical Note. Equation, theorem and table numbers without the prefix "S" refer to the main manuscript. Supplementary Video S1 accompanies this document (Secs. S5 and S7). Records and scripts are available at `https://thanana.github.io/NeuralBeliefMPC.html` (Sec. S7).

Contents

S1 Pendulum protocol **1**

S2 Single-IP correction and budget traces **2**

S3 Controller-first frozen-context study **3**

S4 Pooled frozen-context study (Sec. IV-D) **3**

S5 Closed-loop deployment of the postprocessors **4**

S6 PAM surrogate **5**

S7 Implementation and reproducibility **6**

S1 Pendulum protocol

We use the DeepMind Control Suite cart–pole XML and its official two-pole generator (Tassa et al., 2018), pinned at repository commit `773d5b5d94d8`. These specify model assets; the noisy partial-sensing regulation task is ours, not the suite's reward/reset benchmark. The cart is 1 kg; each pole is a 0.1 kg, 1 m capsule of radius 0.045 m. MuJoCo (Todorov et al., 2012) infers capsule inertias, exported with the compiled masses. Upstream damping, slider range, disabled contacts and actuator settings are retained. RK4 takes two 10 ms steps per 20 ms control interval, and physical force u maps to motor control $u/10$.

States, sensors and profiles. States are $(p, \theta, \dot{p}, \dot{\theta})$ for Single IP and $(p, \theta_1, \theta_{2,r}, \dot{p}, \dot{\theta}_1, \dot{\theta}_{2,r})$ for Double IP, with $\theta_{2,a} = \theta_1 + \theta_{2,r}$. Sensors measure position and absolute angles with independent deviations 0.002 m and 0.2°; velocities are unobserved. Initial positions/relative angles are Gaussian about $(0.02\ \text{m}, 2°)$ or $(0.02\ \text{m}, 2°, -1°)$ with deviations 0.002 m/0.2° and zero velocities; within each seed, methods share the initial draw and noise. Five 20 s profiles are nominal, last-pole mass +50% at 8 s, lower-angle bias ramped to 3° over 8–9 s, all angles missing over 12–12.4 s, and combined. Observers receive masks, not truth or intervention times. Each native simulator step checks $|p| \leq 0.8$ m and each absolute angle within 12°; applied force must stay within 10 N. An episode succeeds only if it completes without any violation; failures are retained.

Controller. Both plants use horizon-50 certainty-equivalent nominal linear MPC with input weight 0.2 and $Q = T_a^T \operatorname{diag}(10, 100\mathbf{1}_{n_p}, 1, 5\mathbf{1}_{n_p})T_a$, where $n_p \in \{1, 2\}$ is the number of poles and T_a maps relative to absolute coordinates. Planning limits are 0.75 m/11°/9.5 N. Terminal sets have 138/312 facets; floating LP support checks give maxima below $1 + 6 \times 10^{-15}$. Invalid plans use shifted feasible plans or saturated LQR recovery, without a safety certificate.

Networks, selection and adaptation. Transition/encoder networks have two width-32 ReLU layers; observation width is 16. Each plant is trained on 64 six-second truth-supervised trajectories (seeds 6001–6064) with 16 validation trajectories (8001–8016). Closed-loop validation uses seeds 8101–8102, nominal/combined profiles and radii $0, 10^{-4}, 10^{-3}, 10^{-2}$; a prespecified lexicographic rule minimizes incomplete runs, violating episodes, then angle-estimation RMSE over the positive radii, giving $10^{-4}/10^{-2}$ for Single/Double IP; for Single IP radius 0 has lower validation RMSE (0.338° versus 0.425°) with equal completion and violation counts. Adaptation uses $\varepsilon_v = 0.002$, $R_v = 0.25$, $B_v = 2$, step 0.02, $\mu = 0.001$, $\beta_J = 10^{-4}$, gradient cap 5, eight backtracking attempts every ten ticks and at most 16 cached pairs. MM variants moment-match an IMM (Blom and Bar-Shalom, 1988) with last-pole mass ratios $\{0.7, 1, 1.5, 2\}$, prior weights $(0.05, 0.85, 0.08, 0.02)$ and mixing $0.99I + 0.01\mathbf{1}\mathbf{1}^T/4$; bias is estimated and the learned transition residual is zero. MM–CF adapts only the encoder; its score uses the fixed trained neural transition/observation surrogate. MM dropout clears replay. MM–CF binding uses coupling budgets $1.1197533 \times 10^{-6}/2.6113003 \times 10^{-5}$, half the median positive accepted coupling at calibration seed 8103, fixed before testing.

Tests and metrics. Seeds 96301–96305 give 300 episodes per plant across twelve variants (five independent seed clusters). Angle-estimation RMSE is $\sqrt{\text{mean}_t\, \text{mean}_j(\widehat{\theta}_{j,a} - \theta_{j,a})^2}$ per episode, in degrees, averaged over seeds; tracking RMSE uses true angles relative to upright. Upward-rounded 100-digit bounds give $L_E \leq 8046.614878/10267.124261$. Maximum finite-budget excess over 300,000/298,754 IP rows is $4.04 \times 10^{-15}/1.02 \times 10^{-13}$.

S2 Single-IP correction and budget traces

Figure S1 shows the Single-IP combined profile for seed 96301 (MM–CF binding and comparison observers). Panel (b) plots the projected correction relative to its radius, the accepted parameter step relative to $\varepsilon_v = 0.002$, and the accepted binding coupling relative to its budget; all remain at or below one, as Theorems 1–2 require.

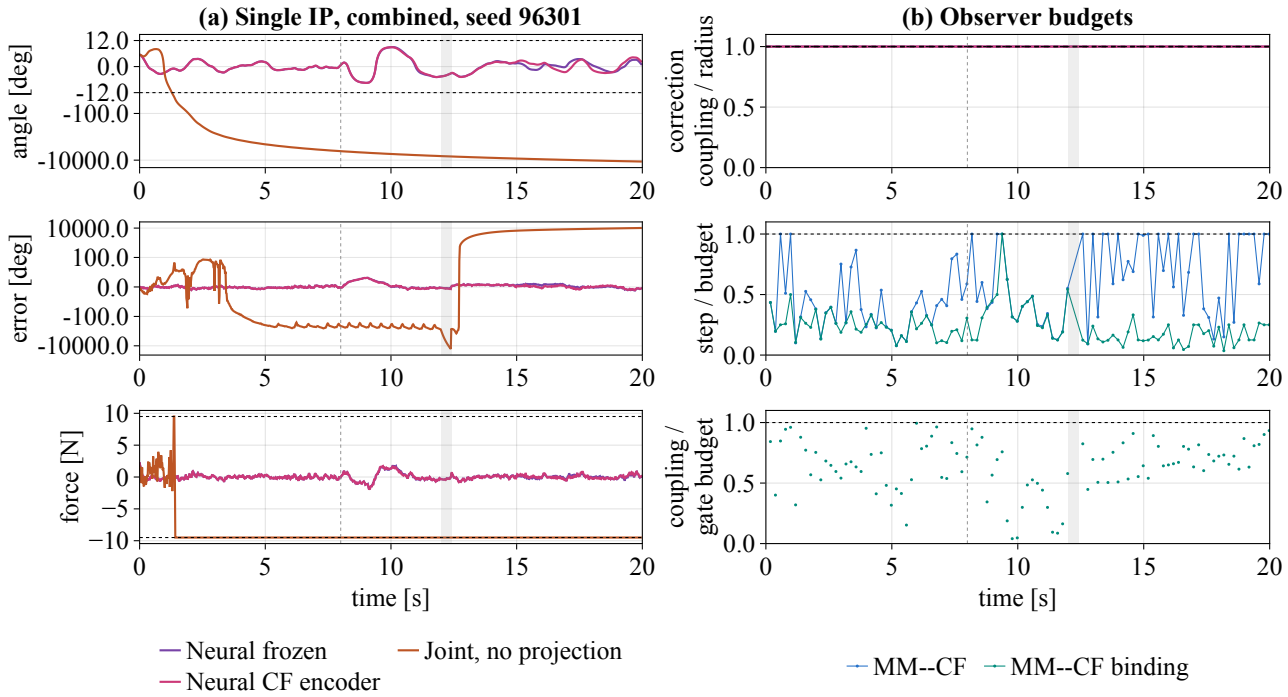


Figure S1: Single-IP combined profile, seed 96301. (a) True angle, estimation error and force; state/error axes are signed-log with physical-value ticks. (b) Projected correction/radius, accepted parameter step/0.002, and accepted binding coupling/budget. Marker: mass/bias onset; shading: dropout. Dashed lines mark physical angle, planning force or observer budgets.

S3 Controller-first frozen-context study

We reuse the 98 combined-seed-96301 contexts per IP; 98/58 Single/Double references lie in the cell. Set $\bar{u}_c \in \{0.001, 0.01, 0.1\}$ N and $\bar{u}_a = 0.1\bar{u}_c$. *Correction-only* uses $\delta_E = \min\{\delta_{\text{nominal}}, 0.99\delta_{\mathcal{C}}, \bar{u}_c/g_M\}$ and $\varepsilon_B = 0.002$; *simultaneous* also caps ε_B at $\bar{u}_a/g_M$; *fixed* uses $\delta_{\text{nominal}} = 10^{-4}/10^{-2}$ and $\varepsilon_B = 0.002$. Identical encoder-gradient proposals use archived frozen caches, norm cap 0.002, scale 0.02 and eight backtracking trials.

Table S1: Input-limit exceedances (correction/update) in the controller-first study: 98 Single/58 Double in-cell contexts per tolerance. C: correction-only; S: simultaneous.

Plant	$\bar{u}_c$ [N]	Fixed	C	S
Single	0.001	37/7	0/0	0/0
	0.01	0/0	0/0	0/0
	0.1	0/0	0/0	0/0
Double	0.001	56/57	0/0	0/0
	0.01	45/41	0/0	0/0
	0.1	0/0	0/0	0/0

All in-cell cases accept updates without cell exits. Correction-only already meets both input limits; simultaneous rejects two additional tight-tolerance Single-IP candidates before accepting smaller steps. Independent reconstruction checks 1,568 candidate scores and 1,764 decisions. Cold-QP checks at archived indices 001, 011, ..., 091 (ten Single, six Double contexts) give 104 distinct solves and 432 point comparisons with cell-law discrepancy below 3.85×10^{-14} N. Median squared factor fractions of reference coupling are 0.609/0.988. These dependent contexts establish neither an adaptation/covariance benefit nor nonlinear safety.

S4 Pooled frozen-context study (Sec. IV-D)

Context generation. The MM–CF binding observer is replayed in all 25 profile/seed test episodes per plant (`revision/julia/replay_seed.jl`), saving every adaptation context exactly as the original `run_guard_replay.jl` does for combined/96301. All 50 replayed episode logs equal the archived test logs in every logged column (maximum absolute difference 0), and the combined/96301 frozen records and raw proposals equal the archived ones bit for bit. `run_study.jl` and `extract_raw.jl` then produce the encoder-gradient proposal, the first cached-score-passing candidate and the raw mean offsets of the old and new parameters for each context.

Problem data. As in the archived study, the incumbent d_0 is the old raw mean offset scaled onto the scalar radius $r_s = \min\{R_E, 0.99r_C, \bar{u}_c/g\}$, the proposal q is the new raw mean offset, $\eta = 0.99$, $\varepsilon_B = \varepsilon_{\text{phys}} = 0.002$ and $\bar{u}_a = 0.1\bar{u}_c$. Cell facets that can cut the radius-R_E ball are retained. The predicted-input map G_U is rebuilt from the stored A, B, K, P, Q and $R = 0.2$ by the pre-stabilized condensing of `mpc_core.jl`; its first row equals $-K_M S_x$, up to the sign convention of the stored gain, within 2×10^{-12}. Row norms peak at $k = 0$ for both plants ($\max_k \|G_k\| = 21.10/79.95$), and $\operatorname{rank}(G_H) = n$ first at $H = 4/6$ (smallest singular values $2.4 \times 10^{-5}/8.0 \times 10^{-8}$, so G_H is numerically near-singular there). For Double IP, $\gamma_1 = 3.92$, $\max_k \gamma_k = 9.77$ at $k = 5$, $\gamma_A = 3.92$, $\|G_A\| = 86.50$ and $|K_M B| = 0.515$; for Single IP, $\gamma_1 = 0.57$, $\max_k \gamma_k = 2.31$ at $k = 10$.

Solvers and certificates. Projections use Clarabel (scaled variables, tolerances 10^{-12}) with constraint violations below 5×10^{-11}. Restricted to combined/96301, the pooled pipeline reproduces the previous $H = 1/H = N$ results within 1.9×10^{-9}, and $H = 1$ reproduces the archived Dykstra distances within 10^{-6}. An independent check on 564 random in-cell contexts (6768 projections) re-solves each problem more tightly, recovers multipliers by NNLS on the near-active set and evaluates the exact Lagrangian dual: certified distance gaps are below 3.3×10^{-11}, except three near-feasible proposals whose directional distances are themselves below 1.1×10^{-5}.

Results by profile. Table S2 breaks down $\bar{u}_c = 0.001$ N by profile; the pooled values for all tolerances are in Table III of the manuscript. Unavailable Double-IP contexts (reference outside the cell) occur mostly after dropout in episodes that later violate limits. The proposal is closer to the true state than the reference in only 35–48% of contexts, which explains why retaining more of it does not improve accuracy.

Table S2: Pooled frozen-context study by profile at $\bar{u}_c = 0.001$ N (25 episodes per plant). Red.: median distortion reduction; Plan 1: largest predicted-input change with $H = 1$ (rounded up); Worse: share of contexts where the directional mean is farther from truth than the scalar mean; Raw<ref: share where the raw proposal is closer to truth than the reference.

Plant	Profile	In-cell	Red. 1 [%]	Red. N [%]	Plan 1 [$\bar{u}_c$]	Worse 1 [%]	Worse N [%]	Raw<ref [%]
Single	nominal	500/500	1.2	1.2	1.00	48	47	46
	mass	500/500	1.1	1.1	1.00	46	46	47
	bias	500/500	1.1	1.0	1.00	45	45	45
	dropout	490/490	1.2	1.2	1.00	49	49	45
	combined	490/490	1.1	1.1	1.00	42	42	48
Double	nominal	497/500	27.4	9.6	18.68	59	58	40
	mass	495/500	26.1	10.0	18.68	59	59	40
	bias	496/500	28.3	9.9	18.68	63	60	35
	dropout	336/490	27.9	9.4	18.68	57	56	42
	combined	329/490	26.1	9.2	18.68	59	62	39

Across all tolerances, Double-IP directional means are farther from truth in 59–64% of contexts, and by episode median in 24–25 of 25 episodes. Single-IP differences are below 6.4×10^{-5} in normalized units; the directional mean is slightly closer in 54% of contexts at 0.001 N and identical at larger tolerances.

Timing. On the archived contexts, Python Dykstra $H = 1$ projections alone (95th percentiles 17.48–20.44 ms) exceed the 20 ms control interval in 4, 4 and 1 of 58 Double contexts.

S5 Closed-loop deployment of the postprocessors

Setup. The MM–CF binding observer of the 600-episode study is run unchanged (adaptation, cached score, coupling gate, IMM bank), except that at every fully observed tick whose reference lies in the cell the deployed physical mean offset is replaced by its projection onto the correction constraints of Sec. III-E at the reference covariance: the scalar ball $\|d\| \le r_s$, or $\{\|d\| \le R_E,\ \bar{M}d \le 0.99s,\ \|G_H d\|_\infty \le \bar{u}_c\}$ with $H = 1$ or $H = N = 50$. No incumbent or update constraint is used in closed loop; nuisance coordinates keep the original correction. Out-of-cell or masked ticks keep the original map. Directional runs leave the cell more often: median Double-IP postprocessed ticks per episode (of about 1000) are 997/987 for the scalar rule, 793/676 for $H = 1$ and 466/518 for $H = N$ ($\bar{u}_c = 0.001/0.01$ N); Single-IP references stay in the cell at every unmasked tick under all three (`post_ticks`, `fallback_ticks` in `revision/analysis/results/closed_loop_results.csv`). The polyhedral projection is computed exactly by least-distance programming (Lawson–Hanson NNLS) and the ball by bisection on its multiplier; a check against Clarabel on 672 frozen contexts gave feasibility 2.7×10^{-15} and distances never larger than Clarabel's. All 25 test episodes per plant (five profiles, seeds 96301–96305) are run for $\bar{u}_c = 0.001$ and 0.01 N and each policy (300 runs, `revision/julia/closed_loop/`).

Table S3: Closed-loop zero-violation episodes by profile (five seeds each), Double IP.

$\bar{u}_c$ [N]	Policy	Nominal	Mass	Bias	Dropout	Combined	Total
0.001	Original radial	5/5	5/5	5/5	1/5	1/5	17/25
	Scalar	5/5	5/5	5/5	0/5	1/5	16/25
	$H = 1$	4/5	1/5	3/5	1/5	0/5	9/25
	$H = N$	0/5	0/5	0/5	1/5	0/5	1/25
0.01	Original radial	5/5	5/5	5/5	1/5	1/5	17/25
	Scalar	5/5	5/5	5/5	1/5	1/5	17/25
	$H = 1$	3/5	1/5	3/5	1/5	0/5	8/25
	$H = N$	1/5	0/5	1/5	0/5	0/5	2/25

Table S4: Median normalized estimation RMSE per state over the first 8 s (or until the first violation) of the five nominal Double-IP episodes, $\bar{u}_c = 0.01$ N.

Deployed mean	p	θ_1	$\theta_{2,r}$	$\dot{p}$	$\dot{\theta}_1$	$\dot{\theta}_{2,r}$
Original radial	0.0016	0.0321	0.0387	0.0138	0.0279	0.0556
Scalar	0.0012	0.0440	0.0504	0.0043	0.0389	0.0743
$H = 1$	0.0045	0.0530	0.0562	0.0474	0.0416	0.0817
$H = N$	0.0061	0.0463	0.0535	0.0534	0.0404	0.0919

Table S5: Closed-loop Single IP, 25 episodes per entry: zero-violation episodes, median tracking and angle-estimation RMSE (deg), and median per-episode median of $\max_k |G_k d|/\bar{u}_c$.

$\bar{u}_c$ [N]	Deployed mean	Zero-viol.	Tracking	Estimation	Plan
0.001	Original radial	25/25	0.343	0.145	–
	Scalar	25/25	0.343	0.148	0.82
	$H = 1$	25/25	0.341	0.151	1.00
	$H = N$	25/25	0.341	0.151	1.00
0.01	Original radial	25/25	0.343	0.145	–
	Scalar	25/25	0.343	0.150	0.16
	$H = 1$	25/25	0.343	0.150	0.16
	$H = N$	25/25	0.343	0.150	0.16

In Single IP all policies keep every episode safe and track as the original map does, consistent with the negligible frozen-context differences (Table S2).

Video. Part 2 of Supplementary Video S1 animates the mass-change episode with seed 96301 at $\bar{u}_c = 0.01$ N for all four deployed means: the original map and the scalar rule stay within limits, whereas $H = 1$ and $H = N$ violate them first at 10.20 s and 11.26 s after the mass change at 8 s.

Interpretation (Double IP). The scalar rule changes the zero-violation outcome of only 3–4 of 25 episodes relative to the original map (it keeps 15 of the 17 originally safe episodes at both tolerances and adds 1–2 others), and its tracking cost comes from its small radius $r_s \le \bar{u}_c/g$. The directional sets keep larger corrections, but the corrections they keep are those the current or predicted inputs barely see (Theorem 3, Proposition 3). Table S4 shows that cart position and velocity are then estimated much worse than under either the scalar rule or the original map, and these episodes later violate the cart or angle limits; the link between the admitted directions and these states is an interpretation, not a proof. With $H = N$, at every tick where the postprocessor is applied, all predicted-input correction changes satisfy the tolerance, yet only 1–2 of 25 episodes stay safe: plan consistency of the belief correction does not imply closed-loop safety. These runs are frozen-hyperparameter tests of one observer; they do not certify the scalar rule either.

S6 PAM surrogate

The surrogate uses the load balance (6), load levels $(M, C, D) = (15, 5, 1.5), (5, 10, 0.2), (12, 8, 1.2)$ and change times 26/60 s from Nuchkrua and Leephakpreeda (2022). Its first transition is a step, not the reported exponential. The reference combines the steps of their Sec. V-A/Fig. 5 with the sinusoid of Sec. V-B/Fig. 6:

$$r(t) = \begin{cases} 0.015, & t < 30, \\ 0.005, & 30 \le t < 40, \\ 0.03 \sin(2\pi(t-40)/6), & t \ge 40, \end{cases} \qquad [r] = \text{m}.$$

This combination and phase are simulation choices. The assumed pressure rate 1.12 is motivated by the reported bandwidth, not an identified transfer function; force/friction, noise, limits and PD control are assumed, and

the reactor, voltage actuation and hardware data are not reproduced. The state is $(q, \dot{q}, p)$ with a position-bias nuisance coordinate; sensors measure (q, p). The implementation discretizes at 20 ms

$$M\ddot{q} + C\dot{q} + Dq = 10(p - 2.5) - 3q - 0.2\tanh(5\dot{q}), \qquad \dot{p} = 1.12(u - p), \tag{S1}$$

with $0 \le u \le 5$ bar and $u = \mathrm{clip}_{[0,5]}(2.5 + [30(r - \widehat{q}) - 35\widehat{\dot{q}}]/10)$. Measurement deviations are 0.5 mm/0.02 bar; process deviations are $0.001\sqrt{0.02}$ m/s and $0.003\sqrt{0.02}$ bar. The combined profile adds the load changes, a 3 mm bias at 35–36 s and dropout at 48–49 s; $|q| \le 6$ cm and $0 \le p \le 5$ bar are checked at sampled ticks. Two fixed width-24 random ReLU layers feed ridge-trained heads; only the encoder bias $\beta_E \in \mathbb{R}^3$ adapts, with $S_x = \mathrm{diag}(0.05, 0.10, 1)$, raw normalized offset $0.2\tanh(W_E\,\text{features} + \beta_E)$, inherited covariance and $W_2 = \rho \le 0.2\|\beta'_E - \beta_E\|$. Validation seeds 8101–8102 select radius 0.005; $\varepsilon_v = 0.015$, $\varepsilon_B = 2 \times 10^{-4}$ and ball/box radii 0.25. Figure S2 repeats Fig. 3 of the manuscript at full width.

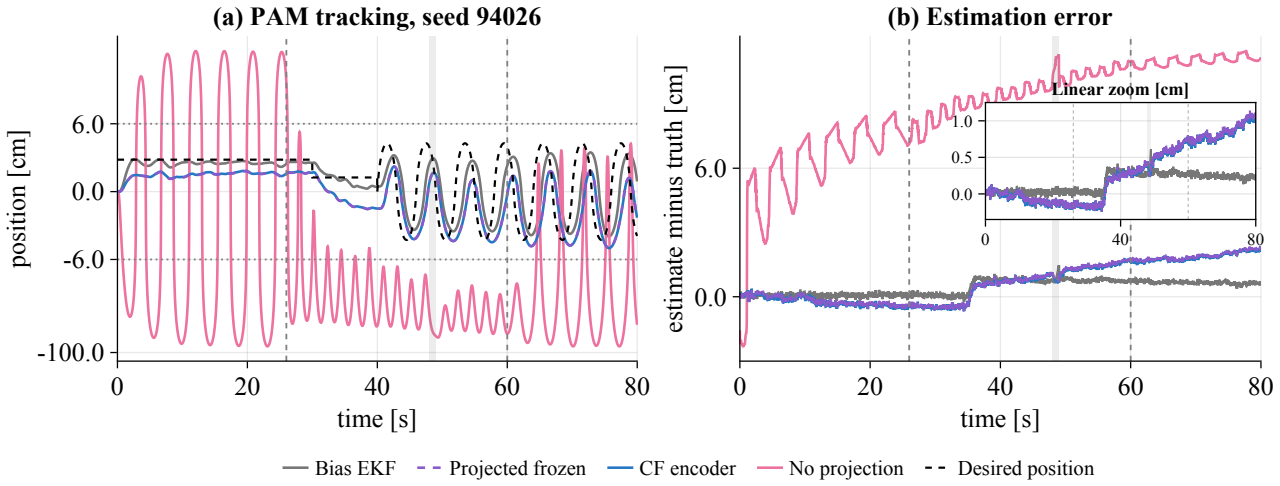


Figure S2: PAM combined seed 94026: tracking and estimation error; signed-log axes with physical-value ticks. Markers: load changes; shading: dropout; dotted: ±6 cm. Inset: linear-scale error (dashed: projected frozen).

S7 Implementation and reproducibility

Code and data. The following material is available at `https://thanana.github.io/NeuralBeliefMPC.html`. An archive contains pinned per-plant Julia environments (`Project.toml`/`Manifest.toml`, Julia 1.12.1), trained text checkpoints, raw records of the 600 IP and 120 PAM episodes, the frozen studies with every proposal and trial score, interval and guard audits, and reproduction commands with hashes. The IP runner skips existing episodes, so fresh execution needs a clean output folder; timing-dependent solver branches may differ in principle, although all 50 replays in Sec. S4 were bitwise identical. Historical PAM raw identity cannot be verified.

Its folder `revision/` contains (see `README.md`):

- `julia/run_all.sh`, `replay_seed.jl`, `run_study.jl`, `extract_raw.jl`: replay of 2 plants × 5 profiles × 5 seeds and frozen-context extraction (about one minute per episode). Usage: `bash run_all.sh`; `SEEDS=96301 PROFILES=combined bash run_all.sh` reproduces the archived study.
- `julia/closed_loop/`: closed-loop deployment of the postprocessors (Sec. S5); `bash run_closed_loop.sh`.
- `analysis/seed_study.py`, `seed_summary.py`, `seed_kkt.py`, `propagation.py`: pooled projection study, summaries, dual certificates and the constants of Proposition 3 (Python with NumPy, CVXPY/Clarabel, SciPy); `analysis/results/` holds every context and every closed-loop run, and `analysis/inputs/` the packed replay contexts.
- `julia/pam_figure/plot_pam_column.jl`, `run_pam_figure.sh`: single-column PAM figure (main-paper Fig. 3) from the archived traces (CairoMakie).

- `video/`: scripts of Supplementary Video S1, which is submitted as a separate file (94 s, 1920×1200): Part 1 copies, frame for frame, the saved Double-IP dropout replay of seed 96301 at 0.5× playback (Fig. 2 of the manuscript); Part 2 animates the closed-loop comparison of Sec. S5 from its saved logs; Part 3 summarizes Tables III–IV. No simulation or training is rerun to make the video (`make_video_v5.py`, `make_cl_frames.py`).